T11: OPTIMIZACION DE EXPRESIONES BOOLEANAS

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1. Completar con los valores en decimal y binario de los mapas siguientes:

	00 $\bar{y}\bar{z}$	01 $\bar{y}z$	11 yz	10 $y\bar{z}$
$\bar{w}\bar{x}$ 00	0000 0	0001 1	0011 3	0010 2
$\bar{w}x$ 01	0100 4	0101 5	0111 7	0110 6
wx 11	1100 12	1101 13	1111 15	1110 14
$w\bar{x}$ 10	1000 8	1001 9	1011 11	1010 10

	00 $\bar{y}\bar{z}$	10 $y\bar{z}$	11 yz	01 $\bar{y}z$
$\bar{w}\bar{x}$ 00	0000 0	0010 2	0011 3	0001 1
$w\bar{x}$ 10	1000 8	1010 10	1011 11	1001 9
wx 11	1100 12	1110 14	1111 15	1101 13
$\bar{w}x$ 01	0100 4	0110 6	0111 7	0101 5

2. Sean las soluciones del ejemplo 2: por min términos y por Max términos. Verificar que son la misma función.

2

$$f(A,B,C) = \sum \min(0,2,3,4,7) = \prod Max\,(1,5,6)$$

Demostración: $F = \bar{B}\bar{C} + \bar{A}B + BC = (B+\bar{C})(\bar{A}+\bar{B}+C)$; $(B+\bar{C})(\bar{A}+\bar{B}+C) = \bar{A}B + BC + \bar{B}\bar{C}$

$F = \bar{B}\bar{C} + \bar{A}B + BC$
$F = B(\bar{A}+C) + \bar{B}\bar{C}$
$F = (\bar{B}\bar{C}+B)(\bar{B}\bar{C}+\bar{A}+C)$
$F = (B+\bar{B})(B+\bar{C})(\bar{A}+\bar{B}+C)(\bar{A}+\bar{C}+C)$
$F = (1)(B+\bar{C})(\bar{A}+\bar{B}+C)(\bar{A}+1)$
$F = (B+\bar{C})(\bar{A}+\bar{B}+C)(\bar{A})$
$F = (B+\bar{C})(\bar{A}+\bar{B}+C)$

$F = (B+\bar{C})(\bar{A}+\bar{B}+C) = \bar{A}B + BC + \bar{B}\bar{C}$
$F = \bar{A}(B+\bar{C}) + \bar{B}(B+\bar{C}) + C(B+\bar{C})$
$F = \bar{A}B + \bar{A}\bar{C} + 0 + \bar{B}\bar{C} + CB + 0$
$F = \bar{A}B + \bar{A}\bar{C} + \bar{B}\bar{C} + CB$
$F = \bar{A}B + BC + \bar{B}\bar{C}$

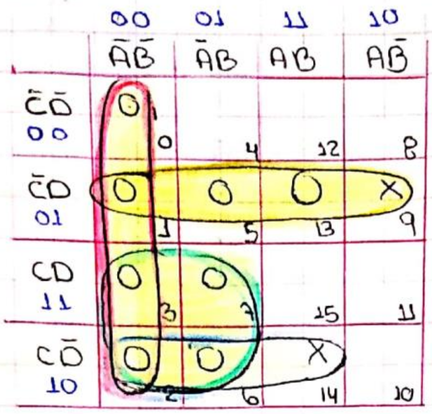
3. Sea la función del ejemplo 3, resolverla por simplificación en Max términos

$$f = \sum min\,(1,3,4,5,7,12,14,15)$$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0 0	 1	 3	0 2
$\bar{A}B$	 4	 5	 7	0 6
AB	 12	0 13	 15	 14
$A\bar{B}$	0 8	0 9	0 11	0 10

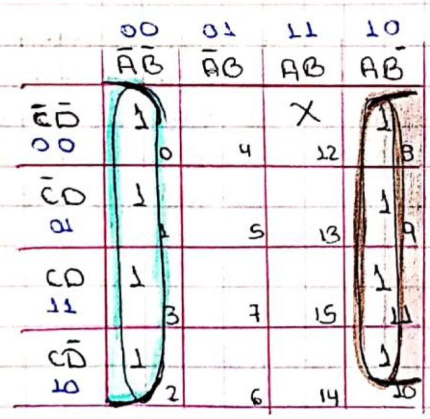
$F = (B+D)(A+\bar{C}+D)(\bar{A}+C+\bar{D})(\bar{A}+B)$

4. Sea la función $G = \sum min\,(4,8,10,11,12,15) + \sum d\,(9,14)$, encontrar su expresión más simplificada expresada en Max términos.



$F = (A+B)(C+\bar{D})(\bar{C}+A)$

5. Sea la función: $G = \prod(4,5,6,7,13,14,15) + \sum d(12)$, encontrar su expresión más simplificada en min términos.



$F = (\bar{A}\bar{B}) + (A\bar{B}) \Rightarrow \bar{B}$

6. Sea la función: $f = B\overline{C}\overline{D} + D(\overline{A} + C) + AC + BCD$, obtener su expresión más pequeña utilizando los mapas k.

$F = B\bar{C}\bar{D} + D(\bar{A}+C) + AC + BCD$

$F = B\bar{C}\bar{D} + \bar{A}D + CD + AC + BCD$

$F = B\bar{C}\bar{D}\cdot 0 + \bar{A}D\cdot 0\,0 + CD\cdot 0\,0 + AC\cdot 00 + BCD\cdot 0$

$F = B\bar{C}\bar{D}(A\cdot\bar{A}) + \bar{A}D(B\cdot\bar{B})(C\cdot\bar{C}) + CD\cdot(A\cdot\bar{A})(B\cdot\bar{B}) + AC(B\cdot\bar{B})(D\cdot\bar{D}) + BCD\cdot(A\cdot\bar{A})$

$F = AB\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}BD(C\cdot\bar{C}) + \bar{A}\bar{B}D(C\cdot\bar{C}) + ACD(B\cdot\bar{B}) + \bar{A}CD(B\cdot\bar{B})$
$ABC(D\cdot\bar{D}) + A\bar{B}C(D\cdot\bar{D}) + ABCD + \bar{A}BCD$

$F = AB\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}BCD + \bar{A}B\bar{C}D + \bar{A}\bar{B}CD + \bar{A}\bar{B}\bar{C}D + ABCD + A\bar{B}CD + \bar{A}BCD$
$\bar{A}\bar{B}CD + ABCD + ABC\bar{D} + A\bar{B}CD + A\bar{B}C\bar{D} + ABCD + \bar{A}BCD$

$A\bar{B}CD$ +	$AB\bar{C}\bar{D}$ +	$\bar{A}B\bar{C}\bar{D}$ +	$\bar{A}BCD$ +	$\bar{A}B\bar{C}D$ +	$\bar{A}\bar{B}CD$ +	$\bar{A}\bar{B}\bar{C}D$ +	$ABCD$ +	$ABC\bar{D}$ +	$A\bar{B}C\bar{D}$
1011	1100	0100	0111	0101	0011	0001	1111	1110	1010
11	12	4	7	5	3	1	15	14	10

$\Sigma min(1, 3, 4, 5, 7, 10, 11, 12, 14, 15)$

	00 $\bar{A}\bar{B}$	01 $\bar{A}B$	11 AB	10 $A\bar{B}$
$\bar{C}\bar{D}$ 00	0	1 4	1 12	8
$\bar{C}D$ 01	1 1	1 5	13	9
CD 11	1 3	1 7	1 15	1 11
$C\bar{D}$ 10	2	6	1 14	1 10

$F = B\bar{C}\bar{D} + \bar{A}D + CD + AC$